

CHUKWUDI MICHAEL OKEREAFOR

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Machine Learning Engineer with a proven track record of designing, deploying, and scaling intelligent systems that drive operational efficiency and user engagement. Expertise in end-to-end ML pipeline development, Generative AI (LLMs, RAG), and building automated data infrastructures. Adept at translating complex business problems into scalable, data-driven products, with strong experience mentoring technical teams and communicating strategic insights to executive leadership.

SKILLS

Machine Learning & AI:	Predictive Modeling, Recommender Systems, Computer Vision (Pose Detection), Generative AI (OpenAI Embeddings, RAG, FAISS), Causal Inference, A/B Testing.
Languages & Frameworks:	Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, JavaScript, TypeScript, C.
Data Engineering & MLOps:	dbt, DuckDB, CI/CD, Model Deployment, Feature Engineering, Data Pipeline Automation.
Cloud & Infrastructure:	AWS, Google Cloud Platform (GCP), Vercel, Render, GitHub, Jira API/Slack API integrations.
Data Visualization:	Power BI, Looker Studio, Google Sheets, Executive Dashboarding, Product Analytics.

WORK EXPERIENCE

Data Scientist & Machine Learning Engineer (Technical Mentor)

ALX Africa

September 2023 - Present | Mauritius (Remote)

- **Scaling ML Products (PeerFinder 3.0):** Engineered and scaled a custom matching application utilizing a Python backend and a TypeScript frontend (deployed via Render and Vercel) that successfully paired 6,174 users with a 99.2% match rate and a median matching time of 29 minutes, driving a 98% user activation rate compared to 50.2% for non-users.
- **Generative AI & LLM Deployment:** Built and deployed an LLM-powered Q&A chatbot utilizing OpenAI embeddings and FAISS vector search, significantly improving response accuracy and reducing query resolution time by 85%.
- **Data Infrastructure & Automation:** Designed a 3-step automated daily workflow (export, import, run) and deployed a Vercel-based Executive KPI Dashboard using conditional formulas and data slicing to track student pacing and course activation metrics, granting leadership real-time visibility into enrollment trajectories and operational targets.
- **Feedback Engine & Analytics Pipeline:** Developed an automated engine to ingest, clean, and visualize large-scale survey and CSAT data across 8 distinct tech programs, replacing hours of manual CSV processing with real-time operational insights.
- **Mentorship & Leadership:** Provide direct technical coaching and mentorship to junior data scientists and learners, increasing program completion rates by 15% while fostering a culture of rigorous data practices.

Data Scientist & Machine Learning Engineer

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December 2022 - August 2024 | CA, United States (Remote)

- **Recommender Systems:** Architected a scalable grocery recommender system for the European market utilizing user preference clustering and historical purchasing data, driving a 20% increase in purchase relevance.
- **Conversational AI:** Developed an AI-driven chatbot designed for empathetic interactions, boosting engagement among vulnerable user demographics by 25%.
- **Computer Vision Solutions:** Designed and trained an AI-powered pose-detection model to track physical movements, resulting in a 30% improvement in the personalization of digital fitness programs.
- **Geospatial Data Science:** Executed complex geospatial data analysis to map urban vulnerability characteristics across South Africa, delivering actionable insights for urban planning initiatives.

Data Analyst

African Women Leaders Think-Tank

October 2013 - December 2016 | Lagos, Nigeria

- **Data Strategy & Cleansing:** Spearheaded end-to-end data extraction and cleansing initiatives, eliminating data silos and contributing to a 30% growth in mentorship program engagement.
- **Business Intelligence:** Conducted rigorous exploratory data analysis to uncover performance trends, directly informing strategic pivoting by the Board of Directors.
- **Reporting:** Engineered comprehensive operational reports and visualization frameworks to track long-term program health and success metrics.

EDUCATION

- **Bachelor of Science (B.Sc), Computer Science** | Delta State University, Nigeria (2015-2019)
- **National Diploma (N.D), Computer Science** | Rivers State Polytechnic, Nigeria (2010-2012)

CERTIFICATIONS & AWARDS

- **Awards:** Medal of Excellence (4th Position), Digital Capabilities for Business (Data Analysis), BRICS Skills Challenge, China (2022).
- **Certifications:** AWS Cloud Practitioner; CS50 (Harvard); Data Science (Explore AI Academy); Data Analysis (Udacity).